Problem 2

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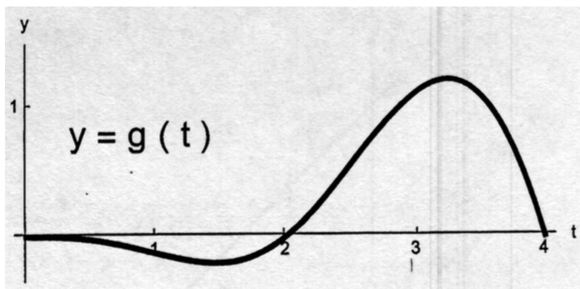
Problem 1 *The graph of g, a continuous function on $[0, 4]$, is shown in the figure. Let $A(x) = \int_0^x g(t)\,dt$, for $0 \le x \le 4$.*



(a) *Circle the correct statement about $A(2)$.*

(i) $A(2) = 0$;

(ii) $A(2) > 0$;

(iii) $A(2) < 0$;

(iv) *none of the previous answers.*

Explanation. Correct choice: (iii) $A(2) < 0$. The curve $g(t)$ on the interval $(0, 2)$ is below the x-axis and therefore the net area is negative.

(b) *Circle the correct statement about $A(3.8)$.*

(i) $A(3.8) = 0$;

(ii) $A(3.8) > 0$;

(iii) $A(3.8) < 0$;

(iv) *none of the previous answers.*

Explanation. Correct choice: (ii) $A(3.8) > 0$. The net area under the curve $g(t)$ on the interval $(2, 3.8)$ is postive and larger than the negative net area on the interval $(0, 2)$. Therefore, the net area on the interval $(0, 3.8)$ is positive.

(c) *Circle the correct statement about $A'(3.8)$.*

(i) $A'(3.8) = 0$;

(ii) $A'(3.8) > 0$;

(iii) $A'(3.8) < 0$;

(iv) *none of the previous answers.*

Explanation. Correct choice: (ii) $A'(3.8) > 0$ because $A'(3.8) = g(3.8)$

(d) *Find the solution of the following initiual value problem:* $y'(x) = g(x)$ *and* $y(0) = 2$.

(i) $y(x) = g(x)$;

(ii) $y(x) = g(x) + 2$;

(iii) $y(x) = A(x)$;

(iv) $y(x) = A(x) + 2$;

(v) $y(x) = g'(x)$;

(vi) $y(x) = g'(x) + 2$;

(vii) *none of the previous answers.*

Explanation. Correct choice: (iv) $y(x) = A(x)+2$. $y'(x) = A'(x) = g(x)$ and $y(0) = A(0) + 2 = 2$

[2]

(e) *Find the expression for* $\displaystyle\int_0^4 |g(t)|\,dt$.

(i) $A(4)$;

(ii) $A(2) - A(4)$;

(iii) $A(4) - A(2)$;

(iv) $A(4) - 2A(2)$;

(v) *none of the previous answers.*

Explanation. Correct choice: (iv) $A(4) - 2A(2)$.

$$\int_0^4 |g(t)|\,dt = \int_2^4 g(t)\,dt - \int_0^2 g(t)\,dt$$

and

$$\int_2^4 g(t)\,dt = \int_0^4 g(t)\,dt - \int_0^2 g(t)\,dt$$

Therefore,

$$\int_0^4 |g(t)|\,dt = \int_0^4 g(t)\,dt - 2\int_0^2 g(t)\,dt = A(4) - 2A(2)$$

(f) *Find the midpoint Riemann sum for the function* g *on the interval* $[0, 4]$ *with* $n = 2$ *(the number of subintervals).*

(i) $g(1) + g(3)$;

(ii) $g(0) + g(2)$;

(iii) $g(2) + g(4)$;

(iv) $2g(1) + 2g(3)$;

(v) $2g(0) + 2g(2)$;

(vi) $2g(2) + 2g(4)$;

(vii) *none of the previous answers.*

Explanation. Correct choice: (iv) $2g(1) + 2g(3)$